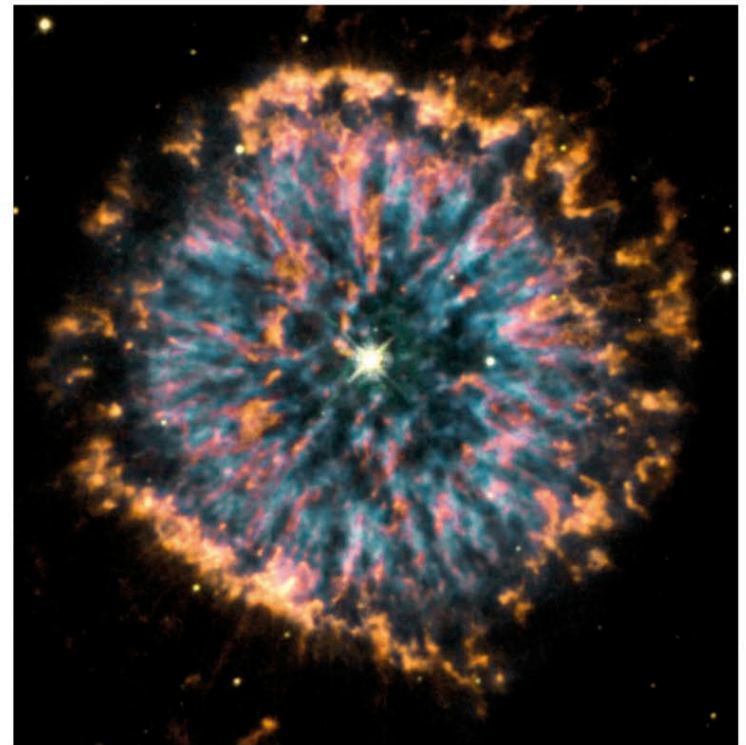


PLANETARY NEBULAE

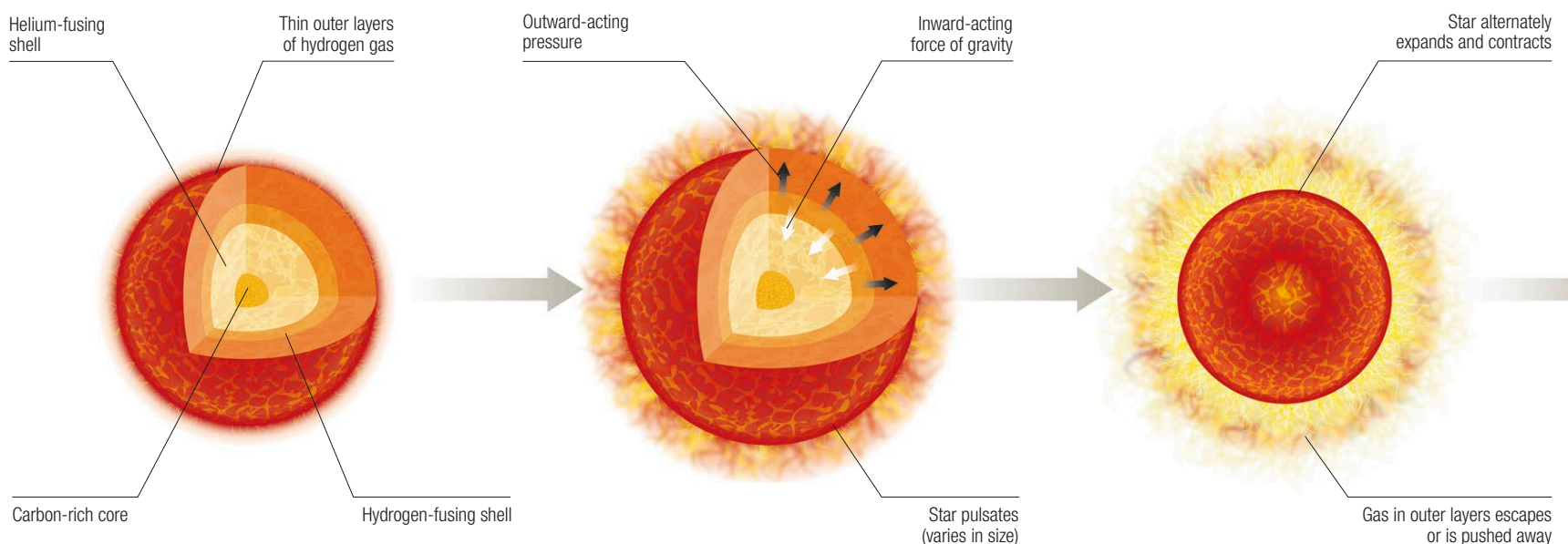
PLANETARY NEBULAE ARE THE HEAVENLY EQUIVALENT OF SMOKE RINGS. RELATIVELY SHORT-LIVED, THEY ARE GRACEFUL CLOUDS OR SHELLS OF GAS PRODUCED DURING THE DYING DAYS OF SUN-SIZED STARS.

Among the finest-looking celestial objects, planetary nebulae have nothing to do with planets—each is just part of the remains of a disintegrated star. The name planetary nebula comes from the nearly spherical, planet-shaped appearance of some of the first of these objects to be spotted. However, modern telescopes have revealed that they actually come in a wide range of shapes. Some planetary nebulae seem to be genuine rings or spherical shells of gas, but others are butterfly-shaped, hourglass-shaped, or can have any of an apparently infinite variety of other complex structures. What all planetary nebulae have in common is that they result from a red giant star becoming unstable at the end of its life and shedding its outer layers. The instability starts when the star begins to run out of materials to fuse in its core (fusion is the joining together of atomic nuclei to make larger nuclei, with the release of energy).



△ Glowing eye nebula

The patterning in this planetary nebula (NGC 6751)—including gas streamers moving away from the bright, central white dwarf—make it look like a giant gleaming eye. Blue regions mark the hottest gas, orange regions the coolest. It is around 0.8 light-years across.



1 Aging red giant

When a star of about the same mass as the Sun nears the end of its life, its energy production rises and it expands into a red giant as its outer layers puff out. An aging red giant has a carbon-rich core surrounded by hot, dense shells of gas where helium and hydrogen fusion occur, producing huge amounts of energy.

2 Star becomes unstable

Two forces maintain the size of the star: inward-acting gravity and outward-acting pressure generated by energy output. The energy-producing fusion reactions are sensitive to changes in temperature and pressure, so tiny variations in these can cause instability in the star's size, leading to large-scale pulsations.

3 Star loses material from outer layers

At the extremes of each pulsation, the red giant expands at such a speed that gas in its outer layers can escape the star's gravity altogether, billowing out into space. The gas is also pushed away by the pressure exerted by particles and photons (tiny packets of light) blasted out from the star's hot core.